CSIRO PUBLISHING

www.publish.csiro.au/journals/ijwf *International Journal of Wildland Fire*, 2004, **13**, 143–156

Slope and wind effects on fire propagation

Domingos X. Viegas

Centro de Estudos sobre Incêndios Florestais–Associação para o Desenvolvimento da Aerodinâmica Industrial, Universidade de Coimbra, Apartado 10131, 3031-601 Coimbra, Portugal. Telephone: +351 239 790732; fax: +351 239 790771; email: xavier.viegas@dem.uc.pt

Abstract. The vectoring of wind and slope effects on a flame front is considered. Mathematical methods for vectoring are presented and compared to results of laboratory experiments. The concept of multiple standard fire spread directions is presented. The experimental laboratory study, included effects of variable wind velocity and direction on point source flame fronts on a 30° inclined fuel bed.

Introduction

It is common to consider wind and topography, together with vegetation as the dominant factors affecting the spread of forest fires. Quite often forest fires occur in areas of complex topography, with mountains and ridges that affect the propagation of the fire in various ways. The presence of wind makes this picture even more complicated with the joint interaction of wind, slope and the fire front. In many instances this is the situation that fire managers and fire fighters have to face.

In order to make the very complex problem of fire behaviour prediction more tractable, the various factors are usually treated separately. The majority of known studies deals with either wind or slope effects separately, and even so they consider only the case in which the rate of spread is aligned with wind velocity or with the local slope gradient. To our knowledge not even the relatively simple case in which wind velocity and slope gradient are aligned is reasonably studied in the literature. A survey of the literature shows that, in spite of its great practical importance for the purpose of fire behaviour prediction, the general case of non-aligned wind and slope has not been studied in a systematic way.

Rothermel and his co-workers (cf. Rothermel 1972) proposed a simple additive model for wind and slope effects both with the same direction and orientation (favourable wind and up-slope propagation), although this situation was not included in their experimental program. In a later work Rothermel (1983) proposed a vector addition of wind and slope effects for the general case of non-parallel wind and slope. Once again it is not apparent that this model was supported on experimental data. This model with variations is extensively used in most fire behaviour prediction systems. Lopes (1994) used a modified version of the additive model proposed by Rothermel. Details on both approaches are given later in the Appendix.

The present research was initially aimed to determine the direction and rate of spread of the head fire front in the case of arbitrary wind and slope. These are necessary inputs for the application of any fire spread model that is based on some contagion algorithm. At an early stage of the experimental tests it was found that the 'head' fire front is not always well defined and it was found also that the application of simple spread models based on the Huygens algorithm (cf. Catchpole *et al*. 1982) cannot describe adequately the observed reality.

Starting from the basic consideration of the existence of two independent convection flows—wind and slope induced buoyancy, respectively—and assuming an additive effect of both flows on the fire front, a simple model based on the vector sum of the corresponding rates of spread is presented and analysed. An experimental device in which wind of arbitrary velocity and direction could blow over a plane surface inclined at a defined angle was used in this study. The spread of a point source fire on a homogeneous fuel-bed composed of dead needles of *Pinus pinaster* was registered with a video camera and the results analysed in order to assess the validity of the concept and the accuracy of the model predictions.

In spite of the limitations of the experimental program developed so far, the results of the present work show some interesting points related to this very important problem of fire behaviour prediction. As will be shown below it is found that the fire front can be decomposed in four main sections that have their own standard fire spread vectors. Each section acts with some independence from its neighbours but it is of course linked to them to form a continuous fire front. This continuity condition applies to all properties of the fire front and leads to the non-uniformity of the fire spread conditions in the general case, as will be shown.

This article is considered as a preliminary study on this topic for the presentation of the problem and of a very

 10.1071/WF03046 1049-8001/04/020143

simple mathematical model that is proposed to solve it. The experimental results, although interesting and original, are still scarce and do not allow the extraction of definitive conclusions about the proposed approach.

Theoretical analysis

In this section a physical analysis of the composite effect of wind- and slope-induced convection on the spread of a section of a flaming fire front is presented. In the discussion emphasis is put on the convection effects although it is recognised that, in this type of front, radiation from the flame or from the reaction zone inside the fuel bed may be the major heat transfer mechanism responsible for the advance of the fire front. The reason for this lies in the fact that convection near the fire front determines the properties of the combustion process and therefore also the shape and size of the flame front and the respective heat transfer fluxes.

Among the various parameters that are suitable for describing the spread of the fire front, its rate of spread $\vec{R}$ is the most widely used. In this article fire spread shall be characterised exclusively by $\vec{R}$. Due to its vectorial nature the rate of spread must be defined both by its magnitude and direction at each point and time step. As we are considering the propagation of a surface fire, $\vec{R}$ must be defined at each point of the fire perimeter characterised by two coordinates x and y. Therefore we must have $\vec{R}(x, y)$. This study is restricted to fire fronts propagating in homogeneous fuel beds on plane surfaces and subject to stationary boundary conditions.

Composite slope and wind effects

It is assumed that the fuel bed is uniform and of constant slope near the section of the fire front that is being considered. It is also assumed that wind velocity vector remains constant during the period of analysis.

The very high gas temperatures that occur near or at the flame front are accompanied by differences of density that induce a flow designated here as natural convection. This natural convection flow caused by buoyancy exists even in fires spreading on horizontal surfaces but it is enhanced by the presence of a slope. In the absence of disturbances like nearby obstacles, other flame fronts, fuel bed discontinuities or edge effects, the flow inside and near the flame front is essentially parallel to a plane defined by the local slope gradient $\vec{S}$ (direction of maximum slope) and by the vertical direction. The OY axis in the reference system considered at the fire front shall be taken as parallel to the maximum slope vector $\vec{S}$ previously defined (see Fig. 1). Let us assume that the overall effect of natural convection on the rate of spread of the fire front can be uniquely related to a rate of spread vector $\vec{R}_s$ that is parallel to the slope gradient. Its magnitude and orientation shall be discussed later.

On the other hand wind can blow from any arbitrary direction in relation to the gradient slope. The convection induced

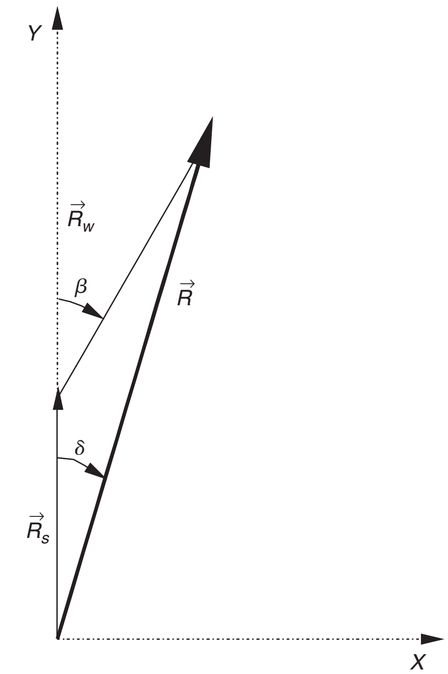

Fig. 1. Schematic view of the wind and slope effects composition, showing the reference system and the vectorial sum of the slope and wind produced rates of spread.

by the wind is designated here as forced convection and its magnitude depends directly on the velocity field near the fuel bed surface. In a similar way we assume that the overall effect of forced convection on the fire front spread can be defined by a vector $\vec{R}_w$ that is parallel to the wind direction, its magnitude and orientation being discussed later.

We designate the angle between wind-induced rate of spread vector and the slope-induced rate of spread vector by β, this angle being measured clockwise. This angle can vary between 0° and 360°. Given the symmetry of the problem in our discussion we consider only positive values of β and in the range $0° < \beta < 180°$. As is shown later, in the same fire perimeter multiple wind and slope induced rates of spread can be defined. In order to avoid confusion the designation of β_0 will be attributed to the angle between the positive wind induced rate of spread and the upslope induced rate of spread.

Considering that $\vec{R}_w$ and $\vec{R}_s$, the vectors representing the rates of spread derived respectively from wind and slope effects, are known and assuming an additive effect of both factors, the local rate of spread due to wind and slope is given by:

$$\vec{R} = \vec{R}_w + \vec{R}_s. \tag{1}$$

In graphical form this equation is represented by the triangle shown in Fig. 1, resulting from the vectorial sum of $\vec{R}_w$ and $\vec{R}_s$. In this figure the angle between $\vec{R}$ and $\vec{R}_s$ is designated by δ and denoted as the deflection (or deviation) angle.

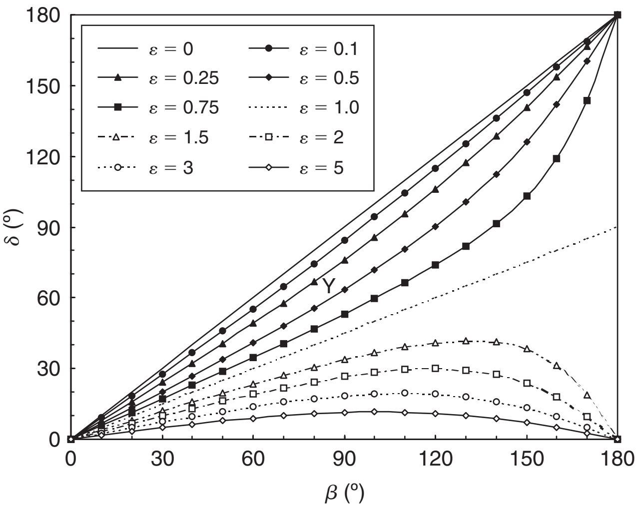

Fig. 2. Variation of deflection angle δ as a function of β, for various values of ε.

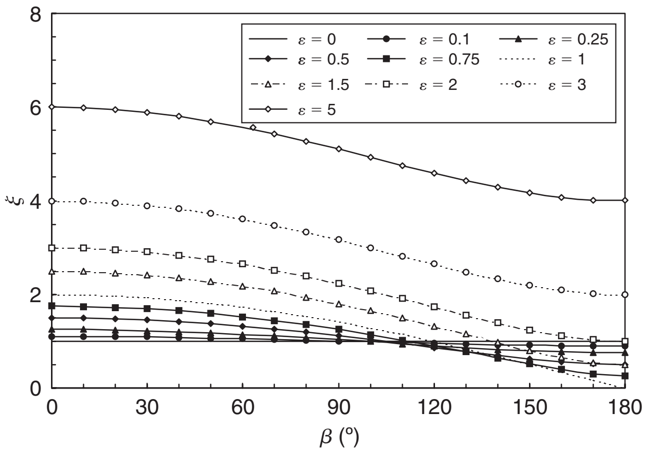

Fig. 3. Variation of the relative rate of spread modulus ξ as a function of β, for various values of ε.

Using the wind induced rate of spread $\vec{R}_w$ as a reference, the following non-dimensional parameters can be defined:

$$\varepsilon = \frac{R_S}{R_W} \tag{2}$$

$$\xi = \frac{R}{R_W}. \tag{3}$$

Using these parameters the following non-dimensional equations can de derived:

$$\tan\delta = \frac{\sin\beta}{\varepsilon + \cos\beta} \tag{4}$$

$$\xi^2 = (\varepsilon + \cos\beta)^2 + \sin^2\beta. \tag{5}$$

The functions described by equations (4) and (5) can be computed easily and their general behaviour is shown in Figs 2 and 3, respectively.

In Fig. 2 the different variation of δ with β depending on whether $\varepsilon < 1$ or $\varepsilon > 1$ can be seen. In the limiting case of $\varepsilon = 1$ the angle δ is equal to $\beta/2$. When $\varepsilon < 1$ the value of δ increases monotonically and tends to 180° as β increases. On the contrary, when $\varepsilon > 1$, δ has a maximum and then tends to zero when β increases.

In Fig. 3 the monotonic decrease of ξ with β can be seen in all cases. For $\varepsilon < 2$ the value of ξ becomes quite small when β approaches 180°.

Non-dimensional forms

Although equations (2) and (3) are already expressed in non-dimensional form, it is convenient to present new non-dimensional forms in order to allow the use of empirical results obtained in slightly different conditions and to provide laws that can be applied to other fuel bed situations.

If we divide both parts of the second member of those equations by R_0, the so-called basic rate of spread—obtained for the same fuel bed in the absence of slope and wind—we have respectively:

$$\varepsilon = \frac{R_S/R_0}{R_W/R_0} = \frac{f_S}{f_W} \tag{6}$$

$$\xi = \frac{R/R_0}{R_W/R_0} = \frac{f}{f_W}. \tag{7}$$

In these equations f_W, f_S and f represent respectively the wind, slope and wind-slope effect on the rate of spread.

For one given fuel bed the non-dimensional functions f_W, f_S should depend respectively only on the wind velocity and on the fuel bed slope angle. It is found however that there may be some dependence on the moisture content and on other uncontrolled fuel bed properties. In order to cope with this variation and also to compare results obtained in laboratory or field experiments in which fuel moisture content is not a fixed parameter, it is more convenient to use the non-dimensional form given by equations (6) and (7).

Curvature effects

Up to now we have been considering infinite linear fire fronts. It is well known that, if the fire line has a finite width or a curvature radius that is of the order of magnitude of the flame length, the spread properties of the fire front are quite different from the previous ones (Weber 1989). In principle the rate of spread is lower in the case of finite width and/or curved flame fronts. As the functions f_W or f_S are different for straight and for curved fire fronts it is necessary to discuss which ones we must consider.

In his report Rothermel (1983) considers only straight fire fronts and therefore no other conditions can be obtained from his fire spread model. On the other hand Rothermel's model is applied extensively to curved fire fronts. This may be acceptable for practical purposes if the curvature radius of the fire front is greater than 5–10 times the flame length, which is the case in many instances.

In the present experiments point ignition fires were used in order to determine the main fire spread direction and the magnitude of the rate of spread vector. Given the scale of the experiments the condition of quasi-straight fire lines—high values of the curvature radius in comparison with the flame length—is not met; therefore, we shall use the functions f_W and f_S corresponding to curved fire fronts obtained in similar conditions, with point ignition fires.

Reference rates of spread

For a given slope angle of the fuel bed we may have either up-slope or down-slope propagating fire. It is well known that the up-slope rate of spread designated as $\vec{R}_{s1}$ is much larger that the down-slope rate of spread $\vec{R}_{s2}$, which is usually of the order of R_0. Depending on fuel-bed properties different authors found that $\vec{R}_{s2}$ can be either slightly higher or lower than R_0 (Fang 1969; Dupuy 1995; Weise and Biging 1996; Viegas 2004). For most practical purposes we can assume, for simplicity, that $\vec{R}_{s2} \approx R_0$.

In order to make our discussion more specific we introduce the following symbols f_{S1} and f_{S2} to designate respectively the multiplying factors to determine the up-slope and down-slope effects respectively, from R_0. The values of these factors must be determined either from models or from experiments. As mentioned above $f_{S2} \approx 1$, for practical purposes. The orientation of $\vec{R}_{s1}$ is that of the slope gradient (pointing towards the top of the slope) and that of $\vec{R}_{s2}$ is the contrary.

Similarly for wind effects we must retain two reference rate of spread values: $\vec{R}_{w1}$ and $\vec{R}_{w2}$, to define respectively the down-wind (favourable wind) or up-wind (contrary wind) propagation conditions. Consequently the symbols f_{W1} and f_{W2} are used to define the respective multiplying factors. Following the considerations that were made above for contrary slope effect we can also assume that $f_{W2} \approx 1$. Similarly the orientation of $\vec{R}_{w1}$ and $\vec{R}_{w2}$ is respectively positive and negative in relation to the wind velocity orientation.

Multiple spread directions

In the case of a fire spreading in a relatively large area we may have sections of the fire front that are more influenced by the local slope and wind conditions than by the presence of another section of the fire. This will occur if the distance between two sections of the fire front—for example one flank and the other—is larger than 5–10 times the flame length. In this case, quite common in real fires, each section of the fire front will act independently from each other and we may find multiple spread directions dictated by local effects. This assertion is not against a previous finding of the

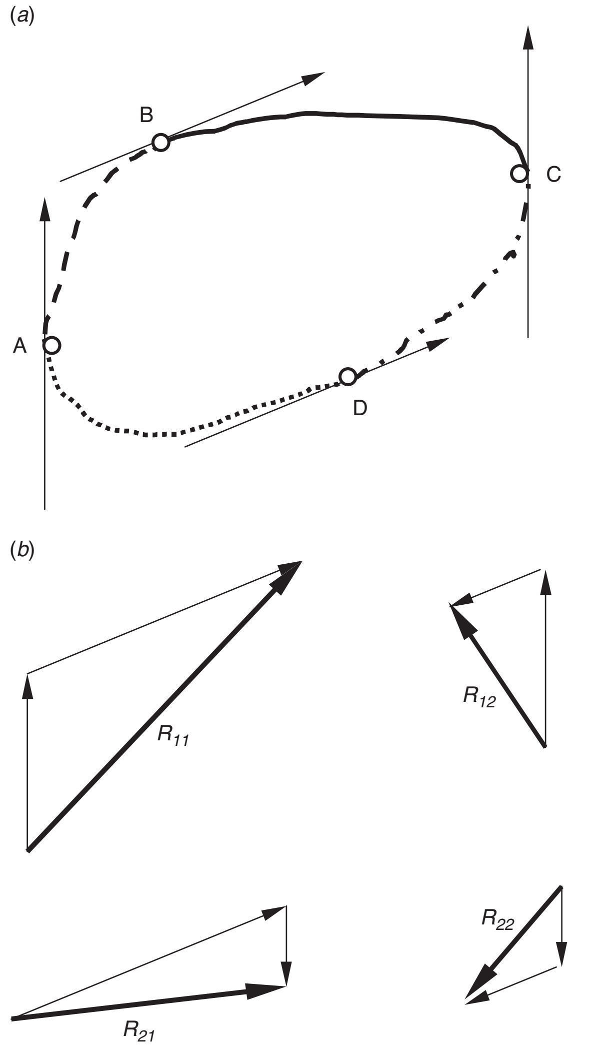

Fig. 4. (*a*) Fire perimeter showing the four sections and their corresponding limiting points. Slope gradient is from bottom to top of figure. Wind direction is indicated by vectors tangent at points B and D. (*b*) Standard spread directions indicated by the dark arrows.

strong interaction between adjacent sections of the fire front as demonstrated in Viegas *et al.* (1994, 1998).

Let us consider again the simpler case of a plane surface fuel bed under uniform and permanent wind conditions. Under these conditions the fire perimeter at any time step will be represented by a convex closed line, i.e. a line without kinks or fingers. The normal vector at any point of this line will always be pointing towards unburned fuel.

If the independent propagation state is reached for given sections of the fire front, we may consider the perimeter of the fire divided into various sections, according to the relative orientation of the wind velocity and slope gradient vectors, as it is shown in Fig. 4*a*. In this figure points *A* and *C* are those tangent to the slope gradient and *B* and *D* are tangent to the wind direction, as shown. Therefore the perimeter of

Table 1. Standard fire spread directions for favourable or unfavourable wind and slope conditions along the fire perimeter

Ref.	Section		Slope	Wind	Standard rate of spread	Non-dimensional factors	
11	*a*	*BC*	Favourable[A]	Favourable	R_{11}	ε_{11}	ξ_{11}
12	*d*	*AB*	Favourable	Contrary	R_{12}	ε_{12}	ξ_{12}
21	*b*	*CD*	Contrary	Favourable	R_{21}	ε_{21}	ξ_{21}
22	*c*	*DA*	Contrary	Contrary	R_{22}	ε_{22}	ξ_{22}

[A]Upslope propagating fire.

the fire is divided in four sections, *a*, *b*, *c* and *d*, which have the following cases of standard local fire spread directions presented in Table 1 and in graphical form in Fig. 4*b*.

It is obvious that when the wind velocity and slope gradient vectors are parallel, $\vec{U}//\vec{S}$, then points A and B coincide and the same happens to points C and D. In this simpler case there are only two standard fire spread directions. In the remainder of this article the more general case of non-parallel wind and slope will be considered.

It may happen that one or more of these four standard spread directions is not realistic in physical terms. Such is certainly the case if a resultant rate of spread vector points towards the already burned area. This case must be looked at more carefully.

If the vector sum of a pair of reference velocities gives a negative result, i.e. a vector pointing towards the already burned fuel, we have to consider two possibilities: either the fire is extinguished or it is still spreading. If the fire is extinguished the rate of spread is null. If the fire is spreading, as the dominant effect is contrary to fire spread, the situation will be similar to a down-slope or contrary wind fire. Consequently the rate of spread of the fire front will be of the order of the basic rate of spread for that fuel-bed, i.e. R_0. The model does not indicate if the fire will be extinguished or if it will propagate in a particular case; this input must come from another source.

Usually the standard spread direction defined by $\vec{R}_{11}$ corresponds to the head of the fire and it is the most relevant in terms of fire spread and hazard. This spread direction shall be referred to from now on as the 'main fire' or 'head fire'. Accordingly $\vec{R}_{22}$ correspond to the 'back fire' and $\vec{R}_{12}$ and $\vec{R}_{21}$ to the 'flank fires'. As will be shown later there are situations in which the maximum rate of spread does not correspond to $\vec{R}_{11}$, but to one of the other standard directions. The designation 'head fire' must then be interpreted adequately.

The meaning of these standard spread directions is the following: each one of the four sections of the fire front will be governed by the corresponding standard rate of spread. This does not mean that all the points of each section are propagating at this rate of spread, of course, but it shows the trend of each section. The various sections of the fire front are linked together and therefore there are some restrictions to the movement of the points of each section. In light of the fire line rotation concept (Viegas *et al.* 1994, 1998), the convective flow in the vicinity of the fire front will transport heat across it and will induce a non-uniformity of the rate of spread vector along the fire line for each section, as was observed for linear fire fronts and for point source fires in a slope. Accordingly one can anticipate that those sections with a flame front inclined towards the unburned fuel due to the transverse transport of heat fire line will rotate and tend to become parallel to the standard rate of spread. This can be observed in the experimental results of Catchpole *et al.* (1982). The 'backward' spreading sections of the fire front will propagate remaining essentially parallel to their previous position, with the restrictions imposed by the linkage to the neighbour sections.

Methodology

Experimental procedure

The experimental program was carried out at the Fire Laboratory of the University of Coimbra using the Combustion Table MC II, described in Viegas *et al.* (1998). This table has a platform of 1.6 × 1.6 m that can be inclined in relation to the horizontal up to 40° by 5° steps. This platform can also be rotated around an axis that is perpendicular to the fuel bed surface creating therefore an arbitrary inclination of the edges of the table.

A flow producing device was attached to one of the edges of the table producing a uniform flow essentially parallel to the table surface and to two of its edges. For a given slope angle α the flow direction could be varied independently from the slope gradient direction, by rotating the table (change of β_0). Both angles α and β_0 could be varied independently at 5° steps in the following ranges, (0–40°) and (0–180°), respectively. Although other values of α were tested, the results reported in this article refer to a slope angle $\alpha = 30°$. The orientation angle β_0 of the wind flow was varied at 30° steps.

The flow velocity can be adjusted at will so a wide range of conditions can be considered in these experiments. Two devices were used in this program. The first was a set of low power axial fans attached to the wall of a box with an exit section of 1.6 × 0.25 m, producing a maximum velocity U_1 of 5 m/s; the second device consisted of a single cross flow

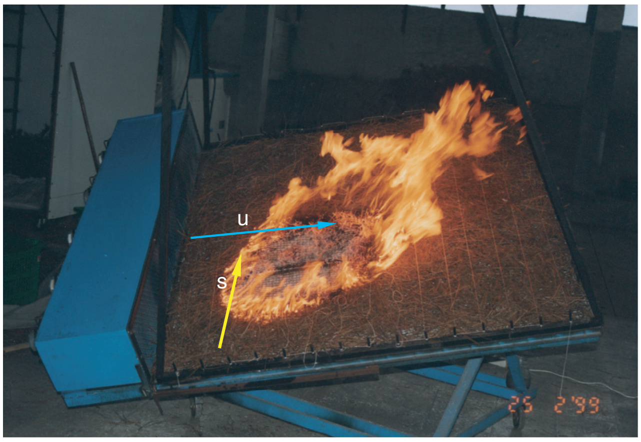

Fig. 5. Combustion Table MC II with the axial flow fans during experiment SW 108 ($\alpha = 30°$: $\beta_0 = 30°$; U_1).

fan with an exit section of 0.9×0.1 m, producing a maximum velocity U_2 of 11 m/s. In a set of preliminary experiments the flow field above the table was measured using a wind vane and it was found that the flow was reasonably symmetrical and uniform for the purpose of the present work. In Fig. 5 the Combustion Table MC II with the axial fan flow producing device is shown.

Although both fan systems could provide a range of flow velocities between zero and the maximum value mentioned above, only the results obtained for the maximum flow velocity values are reported here. In order to avoid damage by the fire front to the fans, the maximum value of β_0 was restricted to 90° and to 150°, respectively for the first (U_1) and the second (U_2) flow producing systems.

The experiments described in this paper were performed in fuel beds made with dead needles of *Pinus pinaster* with a fuel load of 1.0 kg/m^2, on a dry basis. The moisture content of the particles was determined using an electronic moisture balance. This device provided a quick measurement of the fuel moisture in order to correct the fuel load; as it was found that this method is not very accurate, samples of particles were also oven dried for 24 h at 60°C in each test. The present experiments were performed with fuel moisture content (FMC) values in the range of 10–15%, the majority being with $11 < \text{FMC} < 12\%$. The basic rate of spread R_0 was measured several times during the experimental program; its average value was 0.20 cm/s.

In each experiment the fire was ignited at a single point with the assistance of a piece of hydrophyl cotton soaked in a mixture of petrol and kerosene. Some time was left to let the surrounding fuel bed start burning before the fans were switched on. Experiments with a linear ignition line were also performed in this research program. In this case a wool thread soaked in the same mixture was used instead; a single match produced a practically instantaneous and uniform linear starting flame.

A video camera placed above the combustion table with its optical axis always perpendicular to the surface of the table registered the entire sequence of the experiment. The time intervals required by the fire front to cut line threads placed parallel to the *OX* axis at 20 cm intervals were registered during each test with the help of a digital stop watch.

Data analysis

Data analysis was based mainly on the images registered by the video camera placed above the fuel bed. Using a common video acquisition system, images corresponding to pre-selected time steps were digitised to form a file for each test. An example is given in Fig. 6 for the test ref. SW 111, for which $\beta_0 = 90°$; U_1; the time step of each frame is indicated in the figure (see Table 2).

In the sequence of photos shown in this figure it can be seen that the overall behaviour of the fire front changes with time. In the initial stages the fire acts as a whole; after some time the fire front sections are sufficiently separate and begin to act relatively independently from each other, as can be seen at frames of 60″ and 90″ in Fig. 6. The development of the composite effects of wind and slope that were described above can be observed at different sections of the fire perimeter. The top left section of the fire front is dominated by the slope effect whereas the top right section is dominated by the wind effect (cf. frames of 120″, 150″ and 180″ in Fig. 6). This type of behaviour was observed in all experiments. The relatively small size of the combustion table did not allow for an adequate development of the fire front in some cases.

Using standard drawing software the contour of the fire front was drawn point by point for each time step. The whole set of contours was then grouped to form a general picture of the evolution of the fire front in each particular test. An example is shown in Fig. 7 for the same test ref. SW 111.

The main or first standard spread direction, corresponding usually to the head section of the fire front was easy to identify. Therefore the corresponding deviation angle δ_{11} and rate of spread R_{11} were determined graphically from the contour lines.

The identification of the second and third standard spread directions is not so easy as for the head fire; therefore, a different methodology was used to analyse these spread directions. A graphical construction of the vector addition was made in each case in order to determine the respective standard rate of spread direction. A straight line starting from the fire origin was drawn parallel to the standard spread direction and the rate of spread of the fire front along this line was determined. As a result the deviation angle was the same as was given by the model and only the corresponding rate of spread was determined experimentally.

The fire front movement analysis derived from the video camera images was subject to some error as part of the fire front was hidden by the flame. This happened at the head sections of the fire front and might induce some error in the

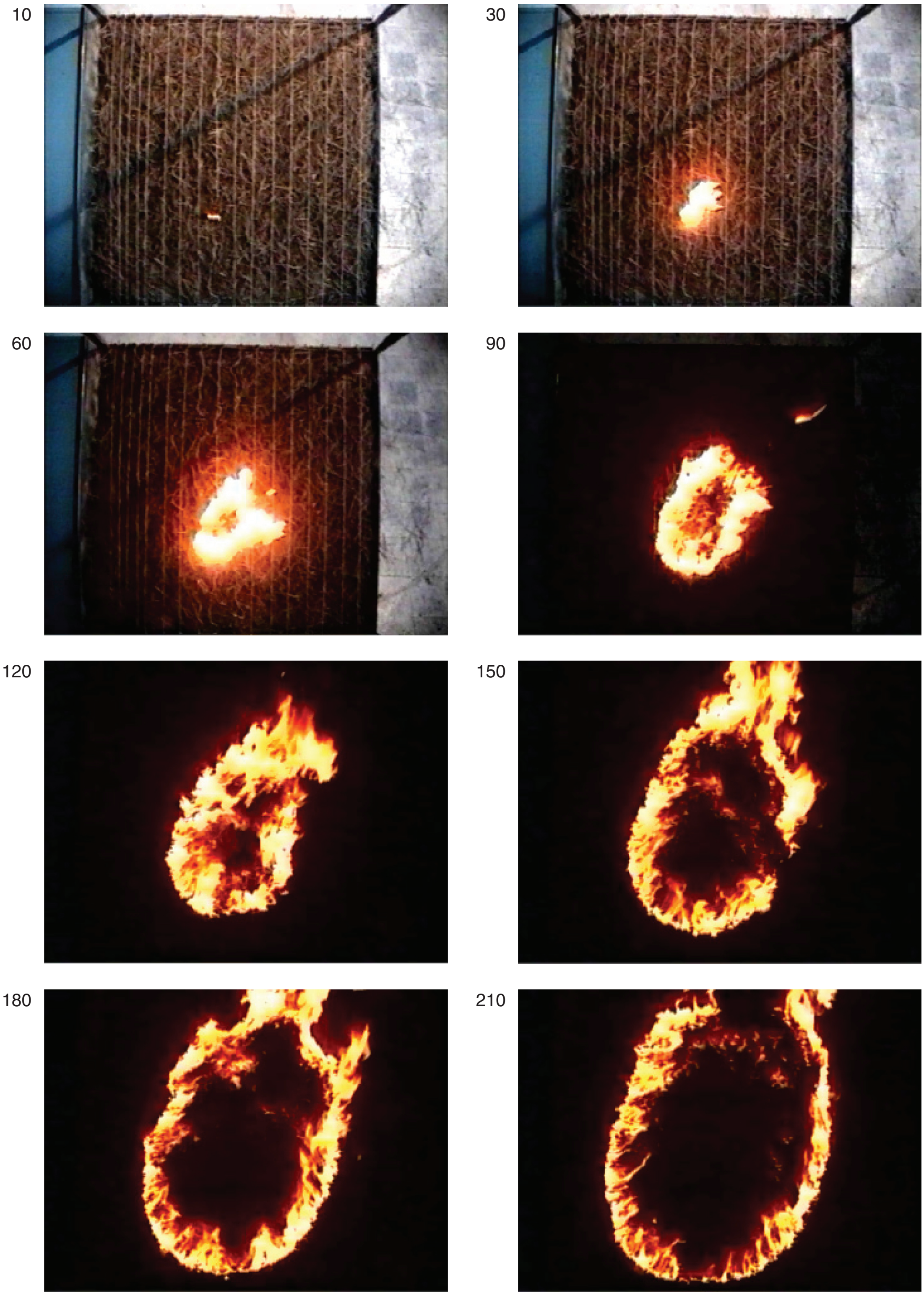

Fig. 6. Sequence of photos taken at different time steps of the test SW 111 ($\alpha = 30°$: $\beta = 90°$; U_1). Wind is blowing from left to right, slope gradient is from bottom to top of the figure. Time from test start is indicated in seconds near each frame.

Table 2. Parameters of two sets of experiments

Wind velocity U_1		Wind velocity U_2	
Ref.	β_0	Ref.	β_0
SW 108	30°	SW 208	0°
SW 110	60°	SW 209	30°
SW 111	90°	SW 210	60°
SW 120	30°	SW 211	90°
SW 124	60°	SW 212	120°
		SW 213	150°

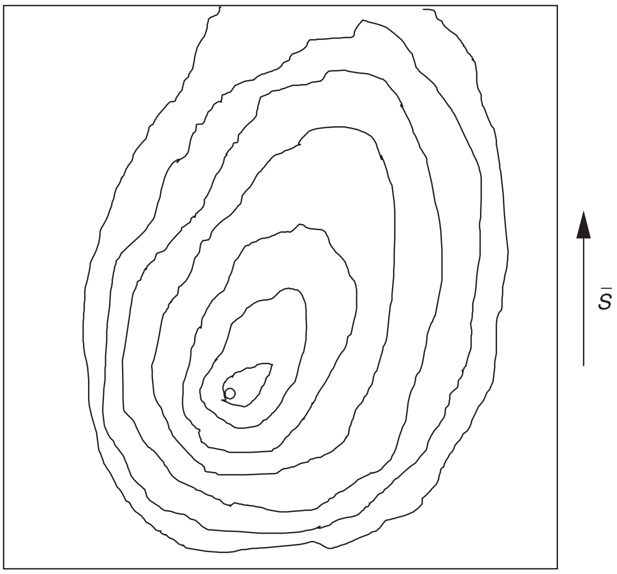

Fig. 7. Schematic representation of the contours of the fire line at different time steps of the test SW 111 ($\alpha = 30°$: $\beta_0 = 90°$; U_1). The black square corresponds to the limit of the fuel bed. The conditions are the same as for the previous figure. Wind is blowing from left to right and the slope gradient direction is shown in the figure.

definition of the exact location of the fire front. For this reason the images were analysed always by the same person using the same evaluation criteria in order to minimise the errors. The author believes that the major conclusions of this paper are not hampered by this fact. In further experiments an infrared camera is being used in order to avoid this source of error.

Test parameters

A very extensive program of experiments was carried out during this study. A detailed list can be found in Pires and Sousa (1999). With the exception of very few exploratory experiments all the others were analysed in detail using the methodology described above.

Due to space limitations only a selection of the results obtained shall be presented in this article. The parameters characterising the selected experiments are given in Table 2. In all these experiments the slope angle was $\alpha = 30°$.

Table 3. Reference non-dimensional parameters of two sets of experiments

Flow device	α	Velocity	f_{s1}	f_{w1}	ε_{11}	ε_{21}	ε_{12}	ε_{22}
1	30°	U_1	4.13	1.38	3.0	0.725	4.13	1
2	30°	U_2	4.13	7.30	0.568	0.138	4.13	1

Results and discussion

Reference conditions

A series of experiments was carried out in the combustion table in order to determine the reference spread conditions, namely R_{w1}, R_{s1} and R_0. From the range of available data only some cases belonging to two sets of experiments will be presented here.

As can be seen in Table 3 the choice of the control parameters in both sets of experiments was such to have values of $\varepsilon_{11} > 1$ in one of them and $\varepsilon_{11} < 1$ in the other. The rate of spread R_{w1} produced by the flow device U_1 was only 38% above the basic rate of spread, while the corresponding value of R_{w1} for flow device U_2 was 7.3 times the basic rate of spread. As a result in the first set of experiments slope was dominating wind for the main or head fire front, while the contrary happened in the second set.

Multiple spread directions

The contour lines corresponding to the following cases: SW 209, 210, 211, 212 and 213 are shown in Figs 8–12, respectively. Time steps between these lines are indicated in the legends. In these figures the head, back and the flank fire propagation directions can be easily identified. Using the known values of f_{w1} and f_{s1} for these cases (cf. Table 3), the corresponding spread vectors are also depicted in these figures. The rate of spread vectors are represented in units of R_0. The graphical scale is the same for all this set of figures. Dotted line vectors represent wind- and slope-induced rates of spread at the corresponding section, while the resulting standard rate of spread is drawn in solid lines.

In Fig. 8 the results for $\beta_0 = 30°$ are shown. The dominating effect of wind is clearly shown in this case, with $\vec{R}_{11}$ being the major value among the standard rates of spread. The evolution of the most advanced point of the contour lines is in accordance with the predicted direction of $\vec{R}_{11}$. The left flank of the fire (top right of the figure) is practically parallel to this vector. It can also be seen that the right flank (lower right of the figure) follows a different direction and tends to become parallel to $\vec{R}_{21}$. The sections corresponding to $\vec{R}_{12}$ (top left) and $\vec{R}_{22}$ (lower left) can also be identified and the general agreement with the model can be assessed in the figure.

In Fig. 9 the case for $\beta_0 = 60°$ is shown. The considerations made for the previous figure can be applied to this case as well. It is worth noting that wind effect appears to dominate the main fire spread after some 40 s after ignition.

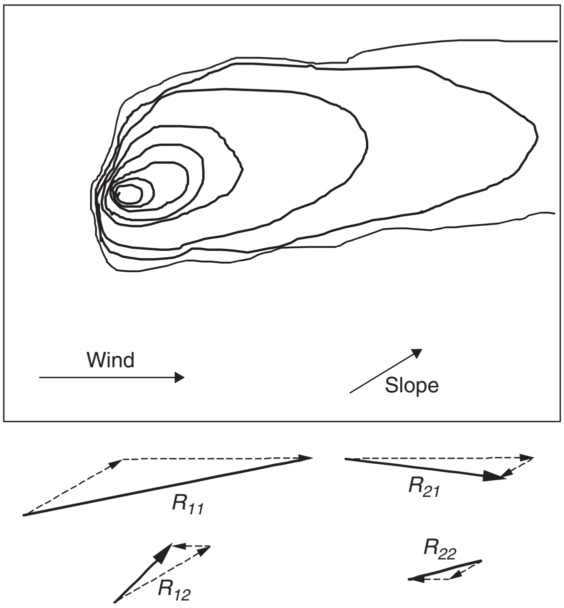

Fig. 8. Fire spread contours for test SW 209 ($\alpha = 30°$: $\beta_0 = 30°$; U_2). Time steps (seconds since fire start) of the contour lines are: 10, 20, 30, 40, 50, 60, 70, 80.

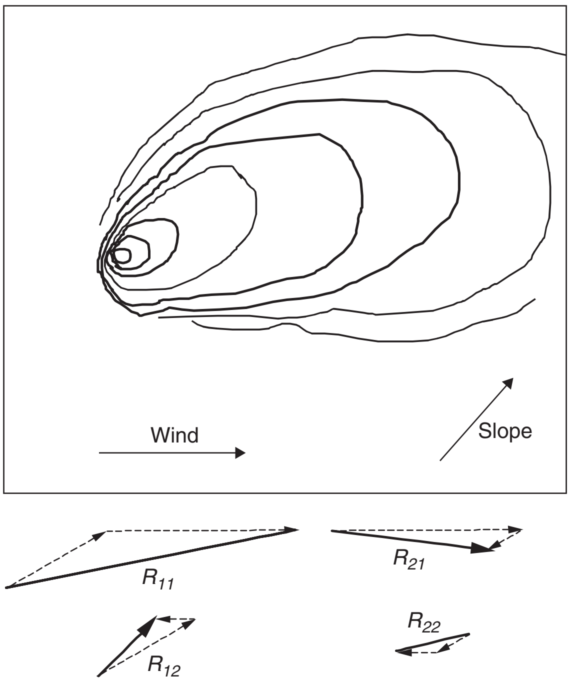

Fig. 9. Fire spread contours for test SW 210 ($\alpha = 30°$: $\beta_0 = 60°$; U_2). The open lines on the left side of the figure correspond to cases in which part of the fire front was extinguished. Time steps (seconds since fire start) of the contour lines are: 10, 20, 30, 40, 50, 60, 70, 80.

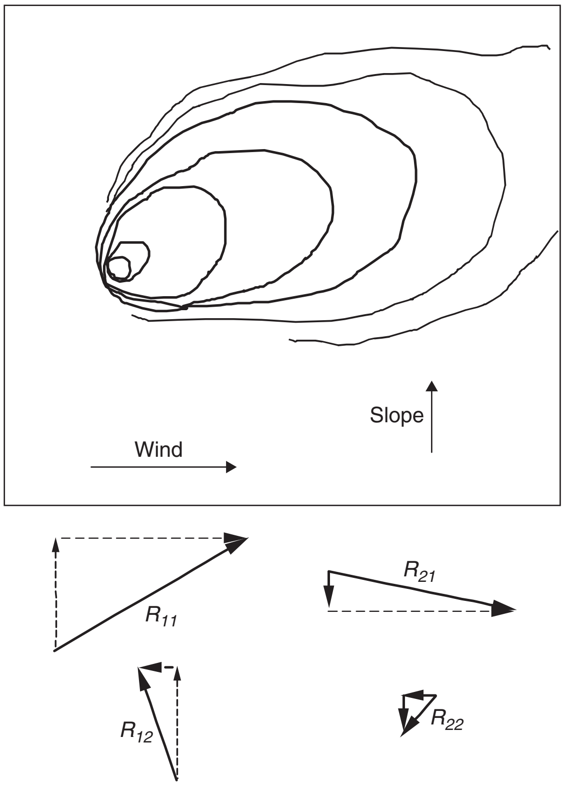

Fig. 10. Fire spread contours for test SW 211 ($\alpha = 30°$: $\beta_0 = 90°$; U_2). Time steps (seconds since fire start) of the contour lines are: 10, 20, 30, 40, 50, 60, 70.

In Fig. 10 wind and slope gradient make an angle $\beta_0 = 90°$. It is interesting to compare the contour lines of Figs 7 and 10, which are obtained for the same values of α and β_0, but with different values of U. The difference between them is of course due to the wind velocity that is much higher in the second case. The section corresponding to $\vec{R}_{12}$ is not formed in Fig. 10. It seems unlikely that the fire line would become parallel to $\vec{R}_{12}$ in this case; at most it should become parallel to the slope gradient.

In Fig. 11 the results for $\beta_0 = 120°$ are shown. In this case $\vec{R}_{11}$ and $\vec{R}_{21}$ are of the same order and indeed in Fig. 11 a sort of bifurcation of the main head can be observed. In this rather interesting case a practically linear fire front is formed at the head of the fire. Size limitations of the combustion table do not allow the verification of the stability of this configuration.

The bifurcation effect mentioned above is shown even more clearly in Fig. 12, for the case of $\beta_0 = 150°$. In this configuration $\vec{R}_{11}$ is smaller than $\vec{R}_{12}$ or $\vec{R}_{21}$, which become the main drivers of two important sections of the fire front. Once again a practically linear fire front is formed in the section BC (cf. Fig. 4) that in this case is practically perpendicular to $\vec{R}_{11}$.

Main spread direction R_{11}

The results obtained for the deflection angle δ_{11} and the fire front spread (R_{11}) of the head fire are analysed in this section, given the relative importance of the head fire for

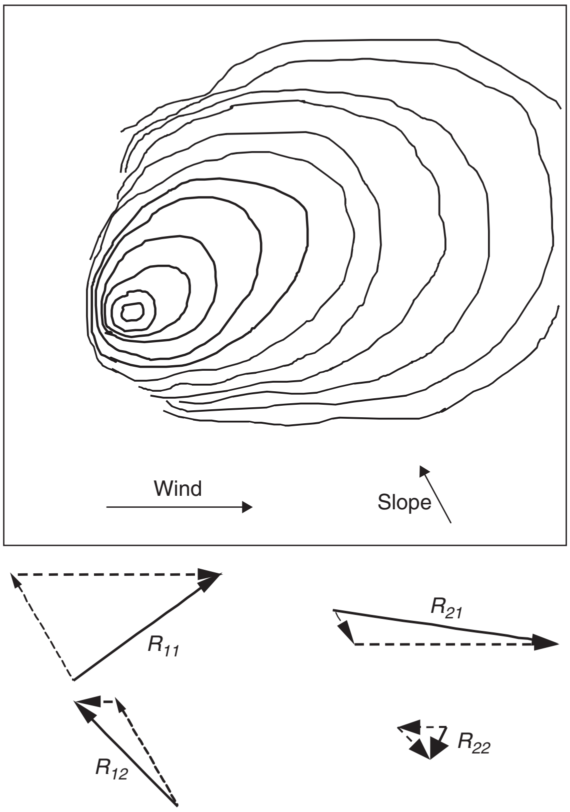

Fig. 11. Fire spread contours for test SW 212 ($\alpha = 30°$: $\beta_0 = 120°$; U_2). Time steps (seconds since fire start) of the contour lines are: 10, 20, 30, 40, 50, 60, 70, 80, 90, 100, 110, 120.

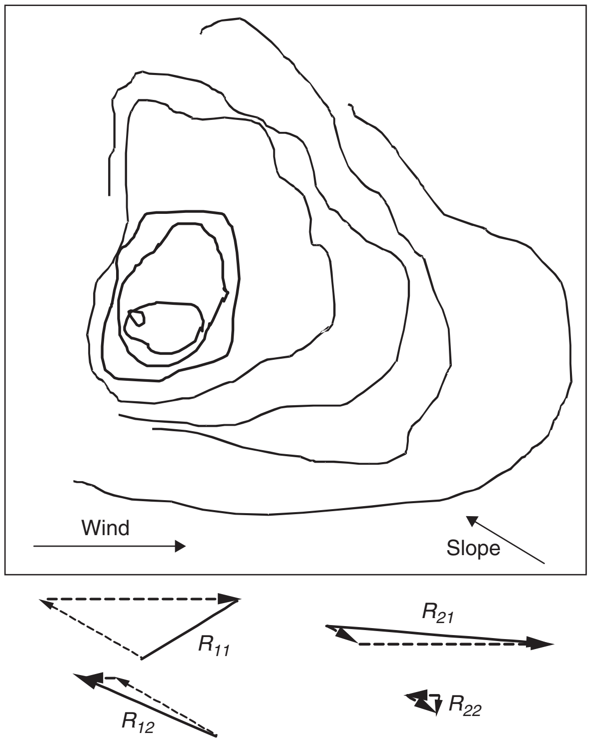

Fig. 12. Fire spread contours for test SW 213 ($\alpha = 30°$: $\beta_0 = 150°$; U_2). Time steps (seconds since fire start) of the contour lines are: 10, 30, 60, 90, 120, 150, 180, 210.

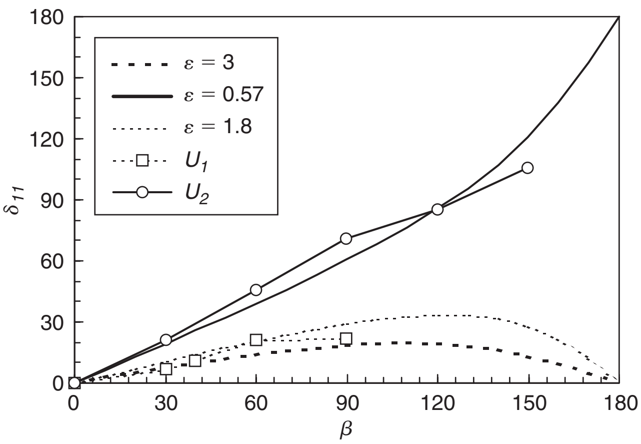

Fig. 13. Deviation angle δ_{11} of R_{11} as a function of β.

practical purposes. The analysis of the other standard fire spread directions is made in the next sections.

The deflection angle δ_{11} of the head fire was determined in each test looking at the fire spread contours searching for the main fire front development. The results are shown in Fig. 13 as a function of β, for the two sets of experiments. In this figure the theoretical lines according to equation (4) are represented for $\varepsilon = 0.57$, $\varepsilon = 1.8$ and $\varepsilon = 3$. Results of the second set of experiments (U_2) fit quite well to the theoretical curve for $\varepsilon = 0.57$. According to Table 3, the results of the first set of experiments should fit to a curve corresponding to $\varepsilon = 3$, as can be seen in Fig. 13.

The results obtained for the non-dimensional rate of spread ξ_{11}, of the head fire front (R_{11}/R_0) are shown in Fig. 14 as a function of β, for the two sets of experiments. Once again the theoretical curve for $\varepsilon = 0.57$ fits reasonably the data from the second set of experiments, with the exception of the first two points. The results from the first set of experiments seem to fit better to a theoretical value of $\varepsilon = 1.8$, instead of the original experimental value of 3. This discrepancy may be explained by a possible inaccuracy in the determination of ε_{11} for this particular set of experiments. In both cases there is a departure from the theoretical predictions for small values of β. It seems that the concept of adding slope and wind induced effects does not work well for small angles of β. The reason for this is not clear at this stage and further experiments must be made under more carefully controlled conditions to verify these laws.

Second standard spread direction R_{12}

Using the methodology described above for the second standard spread direction, the deviation angle is estimated from

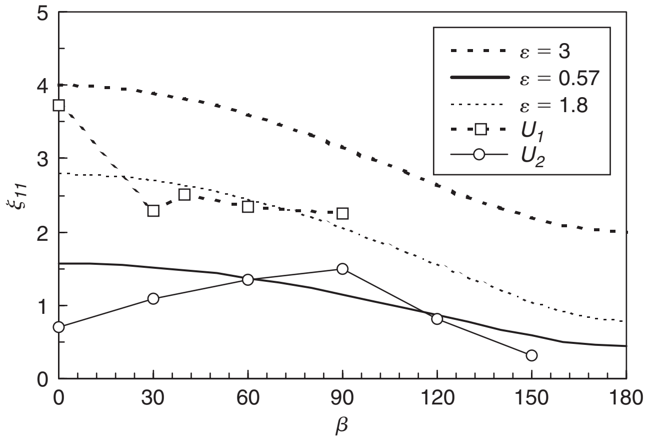

Fig. 14. Non-dimensional rate of spread modulus ξ_{11} of R_{11} as a function of β.

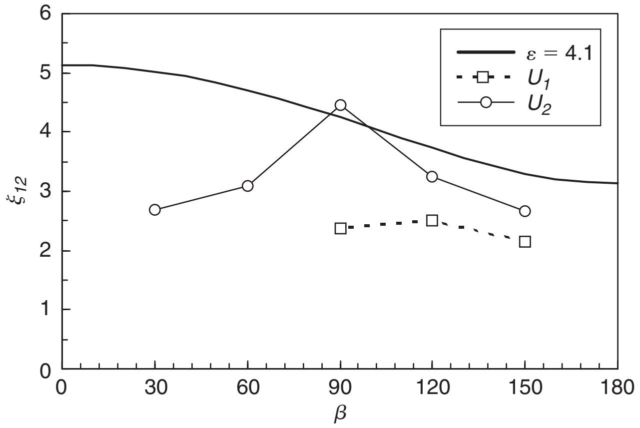

Fig. 16. Non-dimensional rate of spread modulus ξ_{12} of R_{12} as a function of β.

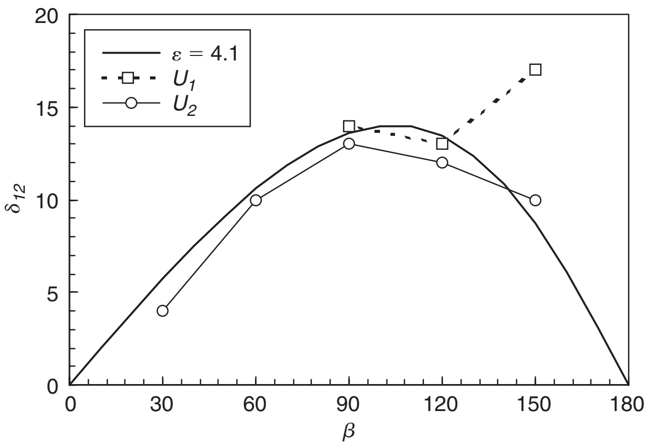

Fig. 15. Deviation angle δ_{12} of R_{12} as a function of β.

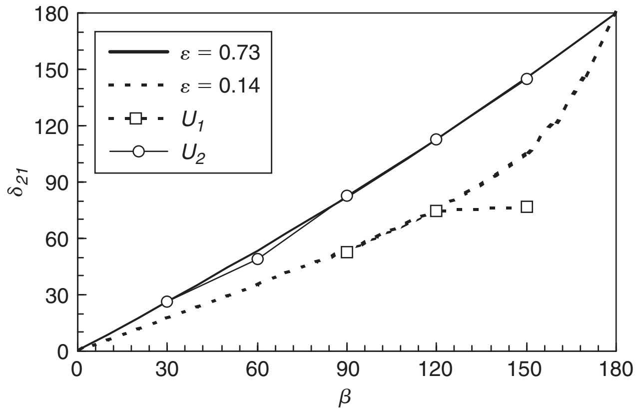

Fig. 17. Deviation angle δ_{21} of R_{21} as a function of β.

the mathematical model and therefore it is practically coincident with the prediction of the model, as can be seen in Fig. 15.

On the contrary the predicted rate of spread is quite different from the model prediction. The experimental results are generally lower than the theoretical values, as can be seen in Fig. 16 where the predicted and measured values of the non-dimensional rate of spread modulus ξ_{12} are shown. The reason for this discrepancy can be found in the concept of fire line rotation proposed in Viegas (2002). According to this concept a section of the fire front that is not perpendicular to the convective flow in its vicinity will not be uniform and its rate of spread will change from one point to another. The fire front movement will be composed by a translation and a rotation that will tend to make that section of the fire front either parallel or perpendicular to the convective flow. As a consequence these sections of the fire front will become flanks that spread with a very low rate of spread, of the order of R_0. In any case the evolution of this section of the fire front is quite complex and it is not completely defined by a single translation rate of spread.

Third standard spread direction R_{21}

Similar comments can be made for the third standard spread direction. The results are shown in Figs 17 and 18 for the deviation angle δ_{21} and for the non-dimensional rate of spread modulus ξ_{21}, respectively.

Rear spread direction R_{22}

The results obtained for the deflection angle δ_{22} and the rate of spread (R_{22}) of the rear or back fire are analysed in this section.

The deflection angle δ_{22} of the back-fire is shown in Fig. 19 as a function of β, for the two sets of experiments. In this figure the theoretical line according to equation (4) is represented for $\varepsilon = 1$. As can be seen the experimental results follow the trend of the theoretical model although the absolute values of δ do not coincide. This is partially due to the

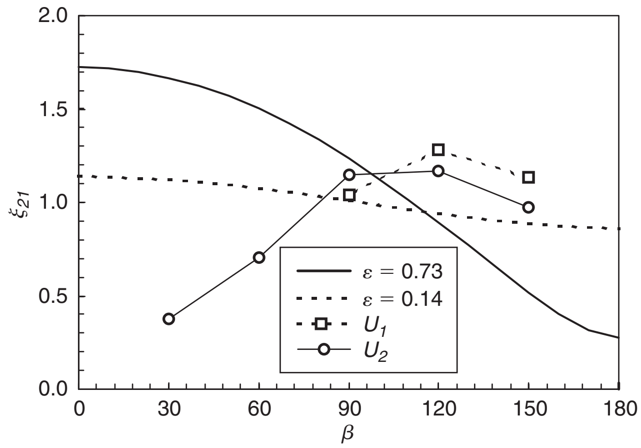

Fig. 18. Non-dimensional rate of spread modulus ξ_{21} of R_{21} as a function of β.

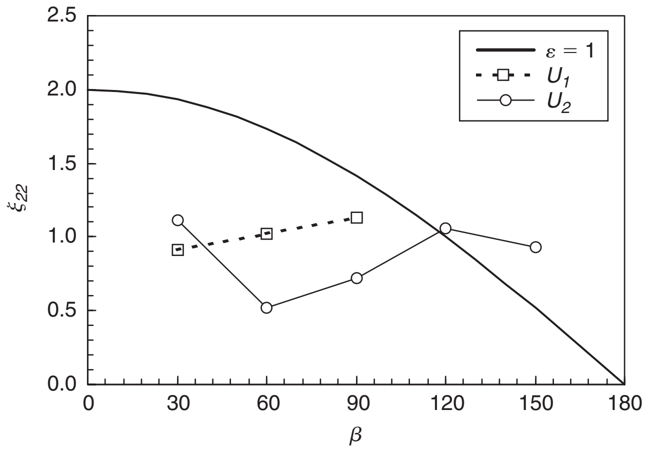

Fig. 20. Non-dimensional rate of spread modulus ξ_{22} of R_{22} as a function of β.

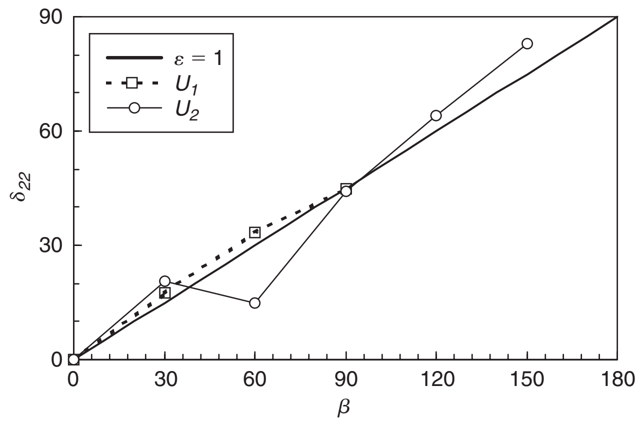

Fig. 19. Deviation angle δ_{22} of R_{22} as a function of β.

fact that it is not easy to determine the direction of R_{22} with accuracy.

The results obtained for the non-dimensional rate of spread ξ_{22}, of the back-fire front (R_{22}/R_0) are shown in Fig. 20 as a function of β, for the two sets of experiments. The experimental values of ξ_{22} are practically constant and they do not follow the trend predicted by the theoretical curve for $\varepsilon = 1$ that is shown in that figure. Our results confirm that, in the presence of wind and slope, the modulus of the rate of spread of the back-fire tends to be of the order of R_0 ($\xi_{22} \approx 1$).

Discussion and conclusion

The theoretical model presented in this article describes the interaction of the convection effects induced by wind and topography in the general case of a surface fire spreading in a uniform slope with arbitrary wind. A vector sum of those effects was presented and discussed.

The existence of four main sections at the fire perimeter when slope gradient and wind velocity are not aligned was shown. These sections behave with a certain degree of independence of each other and are driven by the four standard fire spread directions in the general case, as was shown in this study. The theoretical model based on the additive effect of slope and wind induced convection at the fire front seems to produce reasonably good predictions for the fire front spread direction and at a lesser degree for its rate of spread. Essentially the additive concept proposed by Rothermel (1983) seems to work well, although it is not yet demonstrated which reference values of $\vec{R}_w$ or $\vec{R}_s$ should be taken in each case in order to obtain a good agreement with the experimental results.

As was said above the standard rates of spread do not correspond exactly to the rate of spread at each section of the fire front. Although it seems that in a developed fire these sections may have a situation of quasi-independence in terms of local conditions, due to fireline continuity they are not entirely free to move. Each section must remain 'anchored' by their extremities to their neighbour sections. The concept of fireline rotation brings some light to interpret the evolution of the elements of the various sections of the fireline.

The existence of four standard spread directions in the general case causes a non-symmetric evolution of the fire perimeter shape. This fact implies that the use of simple symmetrical ellipses with their axis parallel to the main spread direction to represent the fire shape can only be an approximation of the actual shape of the fire front.

The effect of the curvature radius of the fire front on the conjugate effect of wind and slope must be analysed more carefully. The present study covers only a very limited range of values of the ratio between the curvature radius of the fire front and the flame front length. Larger scale experiments are required for this purpose.

The present study was also carried out in a homogeneous and plane fuel bed under uniform wind and slope conditions. The real world situation is usually quite different from this. The author believes that a better understanding of the simpler case that is considered here is a first step that will enable a generalisation for the more complex situations of terrain curvature, fuel bed heterogeneity and the existence of non-uniform or non-permanent wind conditions. In a first approximation the case of a large fire in complex terrain can be analysed by sections and for short periods of time for which the simplified conditions considered here may be applied. The existence of multiple fire fronts, fire fingers and other features may be explained under the context of the present research.

This work shall be extended using a larger experimental facility with a better-controlled and more-uniform wind flow. A search of real fires in which the effects of wind and slope are present shall be made. Whenever possible these cases shall be documented and analysed in order to validate the findings of this research. Using the fire line rotation concept and laws a mathematical model for the prediction of the fire front under non-aligned wind and slope shall be attempted.

Acknowledgements

The author acknowledges the help given by his students Mr E. Pires and Mr H. Sousa and by Mr Nuno Luis in the performance of the laboratory experiments and in the preliminary analysis of the data. The author thanks his colleagues Dr A. R. Figueiredo and Dr A. G. Lopes for their critical revision of the manuscript. The comments and suggestions made by anonymous reviewers of this article are also gratefully acknowledged. This work was carried out as part of the work program of the research projects INFLAME and SPREAD supported by the EU respectively under contracts ENV4-CT98–0700 and EVG1-CT-2001–00043. The support given by the Portuguese Science and Technology Foundation to this research program is also acknowledged.

Appendix. Rothermel's formulation and its modified version by Lopes

In this appendix we look at two previous studies on this subject. These are the formulation proposed by Rothermel and its modified version proposed by Lopes (1994).

Rothermel (1972) proposed the following formulation to evaluate the rate of spread R_s and R_w, induced by wind or slope respectively, from the basic rate of spread R_0:

$$R_s = R_0(1 + \phi_s) \tag{A1}$$

$$R_w = R_0(1 + \phi_w). \tag{A2}$$

According to this formulation the modified rate of spread can be considered as the sum of the basic rate of spread and a variation induced either by slope or by wind:

$$R_s = R_0 + \Delta R_s = R_0 + \phi_s R_0 \tag{A3}$$

$$R_w = R_0 + \Delta R_w = R_0 + \phi_w R_0. \tag{A4}$$

For ϕ_w and ϕ_s Rothermel presented analytical equations based on laboratory and on field experiments. The function ϕ_s depended on the slope angle α and on the porosity of the fuel bed; ϕ_w depended on a reference wind velocity and on the surface-to-volume ratio of the particles and on the fuel bed porosity as well. Both functions ϕ_w and ϕ_s are equal to zero when either wind velocity or slope angle is null.

It is easy to see that:

$$f_s = 1 + \phi_s \tag{A5}$$

$$f_w = 1 + \phi_w, \tag{A6}$$

in which f_s and f_w are the corresponding functions defined by equation (6) in the present study.

The above equations (A3) and (A4) were applicable when either wind or slope acted independently from each other. For the case in which wind and slope were present simultaneously, but parallel to each other, Rothermel proposed the following formulation for the rate of spread modulus:

$$R = R_0(1 + \phi_s + \phi_w). \tag{A7}$$

This equation is a generalisation of the previous ones and respects the following condition:

$$U = 0 \quad R = R_s \tag{A8}$$

$$\alpha = 0 \quad R = R_w, \tag{A9}$$

Equation (A7) does not consider that the rate of spread is given by the addition of the elementary rates of spread R_w and R_s. Instead this equation considers the additive effect of the variations ΔR_w and ΔR_s.

In the presentation of the basis of the BEHAVE system, Rothermel (1983) deals with the evaluation of the local spread of sections of the fire front under arbitrary wind and slope conditions. He then proposes a vector sum of the rate of spread $\vec{R}_w$ induced locally by wind and the rate of spread $\vec{R}_s$, induced locally by slope. Although this is not specified by the author, there are indications that the modulus of each one of these vectors is given by equations (A1) and (A2), respectively. Introducing the unit vectors $\vec{e}_w$ and $\vec{e}_s$, defining the slope and wind direction respectively, this formulation is described by:

$$\vec{R} = R_0(1 + \phi_w)\vec{e}_w + R_0(1 + \phi_s)\vec{e}_s. \tag{A10}$$

It is easy to see that, when $\vec{e}_w // \vec{e}_s$, this equation does not give the same result as equation (A7). In order to overcome this discrepancy, Lopes (1994) adopted the following formulation:

$$\vec{R}' = R_0\,\phi_s\vec{e}_s + R_0\,\phi_w\vec{e}_w + R_0\vec{e}_{sw}. \tag{A11}$$

The unit vector $\vec{e}_{sw}$ is parallel to the vector sum of $\Delta\vec{R}_s$ and $\Delta\vec{R}_w$, as is shown in Fig. A1. This formulation is in accordance with the original equation (A7) proposed by Rothermel for the simpler case of parallel wind and slope effects.

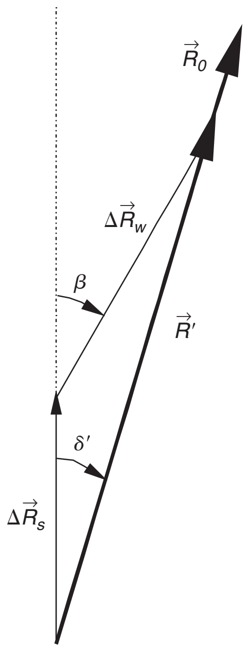

Fig. A1. Vectorial sum of slope and wind effects according to the formulation proposed by Lopes (1994).

Table A1. Evaluation of the composite effect of wind and slope by two formulations ($\beta = 30°$)

ϕ_s	ϕ_w	$\lvert R\rvert/R_0$ (Equation A12)	$\lvert R'\rvert/R_0$ (Equation A13)	δ (Equation A14)	δ' (Equation A15)
3	4	8.70	7.77	16.7°	17.2°
3	10	14.6	13.7	22.1°	23.2°
6	4	11.6	10.7	12.4°	11.9°

It is easy to see that the formulation expressed by equation (A11) is not suitable to be described by non-dimensional equations like those presented above for the initial formulation. Any non-dimensional form of this equation involves explicitly three parameters: β, ϕ_s and ϕ_w. Therefore it is not so convenient for a synthetic analysis. In order to compare both formulations we shall use a non-dimensional form of equations (A10) and (A11) given by:

$$\frac{R}{R_0} = \sqrt{[1 + \phi_s + (1 + \phi_w)\cos\beta]^2 + [(1 + \phi_w)\sin\beta]^2} \tag{A12}$$

$$\frac{R'}{R_0} = 1 + \sqrt{(\varphi_s + \varphi_w \cos\beta)^2 + (\varphi_w \sin\beta)^2}. \tag{A13}$$

It is easy to find that the angles δ between $\vec{R}$ and $\vec{R}_s$, and δ' between $\vec{R}'$ and $\vec{R}_s$ are given respectively by:

$$\tan\delta = \frac{(1 + \varphi_w)\sin\beta}{1 + \varphi_s + (1 + \varphi_w)\cos\beta} \tag{A14}$$

$$\tan\delta' = \frac{\varphi_w \sin\beta}{\varphi_s + \varphi_w \cos\beta}. \tag{A15}$$

It is obvious that equations (A10) and (A11) are not equivalent as is illustrated with a particular case. If we consider $\beta = 30°$, for different values of the pair ϕ_s, ϕ_w, we obtain for each formulation the results that are shown in Table A1. In order to make the comparison easier the modulus of the rate of spread is given with reference to the basic rate of spread R_0.

As can be seen in this table the results given by both formulations for this sample case are quite similar but nevertheless they are slightly different. The accuracy attained in the present experimental program does not permit the extraction of conclusions about the superiority of either formulation. Therefore a detailed discussion of this point is left to a future study.

References

Catchpole EA, de Mestre NJ, Gill AM (1982) Intensity of fire at its perimeter. *Australian Forest Research* **12**, 47–54.

Dupuy JL (1995) Slope and fuel load effects on fire behaviour: laboratory experiments in pine needles fuel beds. *International Journal of Wildland Fire* **5**(3), 153–164.

Fang JB (1969) 'An investigation of the effect of controlled wind on the rate of fire spread.' PhD thesis, University of New Brunswick, Canada. 169 pp.

Lopes AG (1994) 'Modelação numérica e experimental do escoamento turbulento tridimensional em topografia complexa: Aplicação ao caso de um desfiladeiro.' PhD thesis, University of Coimbra, Portugal. 320 pp. [In Portuguese]

Pires EM, Sousa HM (1999) 'Acção do vento e do declive na propagação de uma frente de chamas.' Undergraduate Report, University of Coimbra. 58 pp. [In Portuguese]

Rothermel RC (1972) 'A mathematical model for predicting fire spread in wildland fuels.' USDA Forest Service, Intermountain Forest and Range Experiment Station Research Paper INT-115. 40 pp. (Ogden, UT)

Rothermel RC (1983) 'How to predict the spread and intensity of forest and range fires.' USDA Forest Service, Intermountain Forest and Range Experiment Station General Technical Report INT-143. 161 pp. (Ogden, UT)

Viegas DX (2002) Fire line rotation as a mechanism for fire spread on a uniform slope. *International Journal of Wildland Fire* **11**(1), 11–23. doi:10.1071/WF01049

Viegas DX (2004) On the existence of a steady-state regime for slope and wind driven fire. *International Journal of Wildland Fire* **13**(1), 101–117. doi:10.1071/WF03008

Viegas DX, Varela VM, Borges CM (1994) On the evolution of a linear fire front on a slope. In 'Proceedings of 2nd international conference on forest fire research'. Coimbra, 21–24 November 1994, B.11, pp. 301–318.

Viegas DX, Ribeiro PR, Maricato L (1998) An empirical model for the spread of a fireline inclined in relation to the slope gradient or to wind direction. In 'Proceedings of III international conference on forest fire research, 14th conference on fire and forest meteorology'. Coimbra, 16–20 November 1998, B.05, pp. 325–342.

Weber RO (1989) Analytical models for fire spread due to radiation. *Combustion and Flame* **78**, 398–408. doi:10.1016/0010-2180(89)90027-8

Weise DR, Biging GS (1996) Effects of wind velocity and slope on flame properties. *Canadian Journal of Forest Research* **26**, 1849–1858.